IMPS Platform - Complete Architecture Documentation

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System Overview

IMPS (Imagine. Personalize. Spend time together.) is an AI-powered companion platform requiring specialized architecture for NSFW content delivery, subscription management, and compliance with adult industry regulations.

Target Launch: Q1 2026
MVP Timeline: 3-4 months
Subscription Model: $25/month

Technology Stack

Frontend Layer

- **Framework:** React.js / Next.js (SSR for SEO)
- **Language:** TypeScript
- **Styling:** Tailwind CSS
- **Real-time:** Socket.IO
- **State Management:** Zustand/Redux

Backend Layer

- **API Server:** Node.js with Express/NestJS
- **AI Server:** Python FastAPI
- **Cache:** Redis (session, rate limiting)
- **Session:** JWT tokens (15min access, 7-day refresh)

Database Layer

- **PostgreSQL:** User data, subscriptions, transactions
- **MongoDB:** Chat logs, conversation history
- **Redis:** Sessions, caching, rate limiting

AI/ML Layer

- **Image Generation:** Stable Diffusion XL (self-hosted)
- **Conversational AI:** Llama 3 70B or Mistral Large
- **Vector Database:** Pinecone/Weaviate (memory)
- **Orchestration:** LangChain

Infrastructure

- **Containerization:** Docker + Kubernetes
- **Cloud:** AWS/DigitalOcean (avoid restrictive hosts)
- **CDN:** Cloudflare (DDoS protection, SSL)
- **Storage:** S3-compatible object storage

Payment Processing

Critical Information

⚠ **Cannot use:** Stripe, PayPal, Square, or Razorpay for NSFW content [77][79][82][91]

Primary Payment Gateways

1. CCBill (Recommended Primary)

- **Fees:** 7-10% + $0.30 per transaction
- **Setup:** $750 + $375/year
- **Processing Time:** 3-5 business days
- **Features:** Recurring billing, fraud detection, chargebacks
- **Industry Standard:** Processes $5B+ annually

2. Segpay (Recommended Secondary)

- **Fees:** 5-8% + $0.20 per transaction
- **Setup:** $500-1,000
- **Processing Time:** 5-7 business days
- **Benefits:** Lower fees, good for EU customers

3. Cryptocurrency (Backup)

- **Provider:** NowPayments, CoinGate
- **Fees:** 1-2%
- **Advantages:** No restrictions, global, no chargebacks
- **Disadvantages:** Lower adoption, no recurring billing

Payment Orchestration Strategy

Multi-gateway setup to prevent revenue loss:

```
User Checkout
    ↓
Attempt CCBill (Primary)
    ↓
Success → Activate subscription
    ↓
Fail → Attempt Segpay (Secondary)
    ↓
Success → Activate subscription
    ↓
Fail → Offer Crypto (Tertiary)
```

Subscription Tiers

Free Tier

- Gallery browsing
- 10 messages per day
- Standard responses

Premium Tier ($25/month)

- Unlimited messaging
- Custom AI girlfriend creation
- 50 generated images/month
- Priority support
- Advanced personality customization

Database Schema

PostgreSQL Tables

Users Table

```
CREATE TABLE users (
    user_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    email VARCHAR(255) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    date_of_birth DATE NOT NULL,
    age_verified BOOLEAN DEFAULT FALSE,
    email_verified BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT NOW(),
    last_login TIMESTAMP,
    account_status VARCHAR(50) DEFAULT 'active',
    subscription_tier VARCHAR(50) DEFAULT 'free',
    CONSTRAINT check_age CHECK (date_of_birth &lt;= CURRENT_DATE - INTERVAL '18 years')
);
```

Subscriptions Table

```
CREATE TABLE subscriptions (
    subscription_id UUID PRIMARY KEY,
    user_id UUID REFERENCES users(user_id),
    plan_type VARCHAR(50) NOT NULL,
    status VARCHAR(50) NOT NULL DEFAULT 'active',
    current_period_start TIMESTAMP NOT NULL,
    current_period_end TIMESTAMP NOT NULL,
    payment_method VARCHAR(100),
    amount_cents INTEGER NOT NULL,
    currency VARCHAR(3) DEFAULT 'USD',
    auto_renew BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT NOW()
);
```

Transactions Table

```
CREATE TABLE transactions (
    transaction_id UUID PRIMARY KEY,
    user_id UUID REFERENCES users(user_id),
    subscription_id UUID REFERENCES subscriptions(subscription_id),
    amount_cents INTEGER NOT NULL,
    currency VARCHAR(3) DEFAULT 'USD',
    status VARCHAR(50) NOT NULL,
    payment_gateway VARCHAR(100) NOT NULL,
    gateway_transaction_id VARCHAR(255),
    created_at TIMESTAMP DEFAULT NOW()
);
```

AI Companions Table

```
CREATE TABLE ai_companions (
    companion_id UUID PRIMARY KEY,
    user_id UUID REFERENCES users(user_id),
    name VARCHAR(100) NOT NULL,
    physical_attributes JSONB NOT NULL,
    personality_traits JSONB NOT NULL,
    profile_image_url TEXT,
    created_at TIMESTAMP DEFAULT NOW()
);
```

MongoDB Collections

Conversations Collection

```
{
  _id: ObjectId,
  conversation_id: UUID,
  user_id: UUID,
  companion_id: UUID,
  messages: [
    {
      role: "user" | "assistant",
      content: String,
      timestamp: Date,
      metadata: {}
    }
  ],
  created_at: Date
}
```

Redis Data Structures

Session Storage

Key: `session:{session_id}`
TTL: 24 hours

Rate Limiting

Key: `ratelimit:{user_id}:{endpoint}`
Type: Counter with 60s TTL

Image Generation Queue

Key: `queue:image_generation`
Type: List (FIFO)

AI Architecture

Image Generation System

Technology

- **Model:** Stable Diffusion XL (SDXL) or SDXL Turbo
- **Hardware:** NVIDIA A100 40GB-80GB (2-4 GPUs)
- **Custom Models:** LoRA for character consistency

Image Generation Flow

```
User Request
    ↓
Queue System
    ↓
GPU Processing (5-15 seconds)
    ↓
Content Filter
    ↓
S3 Storage
    ↓
Serve to User
```

Monthly Costs

- GPU Server: $500-1,000
- Storage: $50-100
- Bandwidth: $100-200
- **Total:** $650-1,300/month

Conversational AI System

LLM Selection

- **Llama 3 70B** (uncensored variant) OR
- **Mistral Large** OR
- **Custom fine-tuned model**

Conversation Architecture

```
User Message
    ↓
Intent Classification
    ↓
Context Retrieval (Vector DB)
    ↓
LLM Generation
    ↓
Safety Filter
    ↓
Response
```

Memory System

- **Short-term:** Redis (current session)
- **Long-term:** Vector database (Pinecone/Weaviate)
- **Stores:** User preferences, conversation topics, personality traits

Launch Roadmap

Pre-Launch Phase (Week 1-2)

Week 1: Business & Legal Setup

- [ ] Register LLC or Corporation
- [ ] Obtain EIN / Tax ID
- [ ] Open business bank account
- [ ] Register domain: imps.ai
- [ ] Apply for CCBill merchant account
- [ ] Draft legal documents (ToS, Privacy Policy)

Cost: $2,000-4,000

Week 2: Infrastructure Setup

- [ ] Create AWS/DigitalOcean account
- [ ] Set up VPC and security groups
- [ ] Configure Cloudflare CDN
- [ ] Provision databases (PostgreSQL, MongoDB, Redis)
- [ ] Configure monitoring (Grafana, Sentry)

Cost: $500-1,000 initial + $300-500/month

Phase 1: MVP Development (Months 1-3)

Month 1: Core Backend & Frontend

- [ ] User authentication system
- [ ] Age verification (18+ gate)
- [ ] Database schemas
- [ ] Landing page with gallery
- [ ] Responsive design

Month 2: AI Integration

- [ ] Llama 3 integration
- [ ] Chat interface with WebSocket
- [ ] Conversation history
- [ ] 50+ AI girlfriend gallery
- [ ] Safety content filters

Month 3: Payment & Subscription

- [ ] CCBill payment integration
- [ ] Subscription management
- [ ] Usage quotas (Free: 10 msg/day, Premium: unlimited)
- [ ] Webhook handlers
- [ ] Admin dashboard

Development Cost: $10,000-20,000 (if outsourcing)

Phase 2: Beta Testing (Month 4)

Week 1: Private Beta Launch

- [ ] Invite 50-100 beta testers
- [ ] Set up feedback collection
- [ ] Monitor errors closely

Week 2-3: Refinement

- [ ] Fix critical bugs
- [ ] Improve UI/UX
- [ ] Optimize performance
- [ ] Payment flow testing

Week 4: Pre-Launch Preparation

- [ ] Create social media accounts
- [ ] Design launch graphics
- [ ] Prepare Product Hunt launch
- [ ] Final security audit

Cost: $2,000-5,000

Phase 3: Public Launch (Month 4, Week 4)

Launch Day

- [ ] Deploy production version
- [ ] Post on Product Hunt
- [ ] Share on social media
- [ ] Monitor signups and errors
- [ ] Customer support ready

Phase 4: Growth (Months 5-12)

Month 5-6: Image Generation

- [ ] Set up GPU servers (NVIDIA A100)
- [ ] Integrate Stable Diffusion XL
- [ ] Build customization wizard
- [ ] Image generation queue

Cost: $500-1,000/month (GPU servers)

Month 7-8: Advanced Features

- [ ] Voice messages (text-to-speech)
- [ ] Memory system (vector database)
- [ ] Mobile app (React Native)

Month 9-10: Scaling

- [ ] Auto-scaling infrastructure
- [ ] Database read replicas
- [ ] SEO optimization
- [ ] Paid ads (Google, Reddit)

Month 11-12: Internationalization

- [ ] Multi-language support (Spanish, French, Japanese)
- [ ] EU compliance (GDPR)
- [ ] Multi-currency support

Cost Analysis

One-Time Costs

Item	Cost
Business registration	$500-1,000
Legal documents	$1,000-2,000
CCBill setup	$750
Initial infrastructure	$500-1,000
Marketing launch	$1,000-3,000
Total	**$3,750-7,750**

Monthly Operating Costs

Infrastructure (Minimal Setup)

Component	Monthly Cost
Web/API servers	$300-500
GPU servers	$500-1,000
Databases	$200-400
Redis	$100-200
Storage + CDN	$150-300
Subtotal	**$1,250-2,400**

Services

Service	Monthly Cost
Payment processing	8% of revenue
Age verification	$0.10-0.50/user
Email/SMS	$100-200
Subtotal	**$100-200 + 8% revenue**

Development Team (Optional)

Role	Monthly Cost
Backend developer	$5,000-8,000
Frontend developer	$4,000-6,000
AI/ML engineer	$6,000-10,000
DevOps	$4,000-6,000
Subtotal	**$19,000-30,000**

Revenue Projections

Assumptions

- Subscription: $25/month
- Payment processor fee: 8%
- Conversion rate: 2% (free to paid)

Growth Scenarios

Users	Monthly Revenue	Processing Fees (8%)	Net Revenue	Profit (after $3k costs)
100	$2,500	$200	$2,300	-$700
133	$3,325	$266	$3,059	**Breakeven**
500	$12,500	$1,000	$11,500	$8,500
1,000	$25,000	$2,000	$23,000	$20,000
5,000	$125,000	$10,000	$115,000	$112,000

Breakeven Analysis

- **Minimal Setup ($3k/month):** 133 paying users
- **Full Team ($32k/month):** 1,422 paying users

Compliance & Legal Requirements

Age Verification (MANDATORY)

Implementation [66][83]

- Age gate on landing page (18+ confirmation)
- Date of birth collection and verification
- Email verification required
- Credit card verification (billing address confirms 18+)
- Optional: Third-party age verification (AgeChecker.Net, Yoti)

Content Moderation System

Real-time Moderation [66][76]

- AI keyword detection (block illegal terms)
- Image analysis for prohibited content
- User reporting system
- Manual review queue for flagged content

Prohibited Content

- Underage content (any suggestion)
- Bestiality
- Extreme violence
- Non-consensual scenarios
- Incest themes
- Real person impersonation without consent

Data Privacy & Security

Compliance Standards [66][83]

- **GDPR** (EU users) - Right to deletion, portability
- **CCPA** (California) - Privacy notices, opt-out rights
- **PCI DSS Level 1** (payment data)

Security Measures

- End-to-end encryption for messages
- Tokenized payment storage
- Encrypted conversations (auto-delete after 30 days)
- SSL/TLS for all connections
- Rate limiting to prevent abuse

Required Legal Documents

- Terms of Service (clear adult content warning)
- Privacy Policy (data collection, retention, deletion)
- Cookie Policy
- Content Guidelines
- Refund Policy (7-day money-back guarantee)
- 18 U.S.C. § 2257 compliance statement

Security Architecture

Authentication & Authorization

Auth Flow

- JWT tokens (access: 15min, refresh: 7 days)
- Refresh token rotation
- Device fingerprinting
- Optional 2FA (email/SMS)

Rate Limiting

- Login attempts: 5 per 15 minutes
- API calls: 100/min (free), 1000/min (premium)
- Image generation: 50 per day (premium)

Content Security

Image Storage

- Encrypted at rest (AES-256)
- Signed URLs with expiration (1 hour)
- Optional watermarking (user-specific)

Chat Security

- End-to-end encryption option
- Auto-delete conversations after 30 days
- User-controlled data deletion

Deployment Architecture

Production Environment

```
Production Stack:
├── Cloudflare (CDN, DDoS protection)
├── Load Balancer (AWS ALB)
├── Web Servers (3+ instances, auto-scaling)
├── API Servers (5+ instances, auto-scaling)
├── AI Inference Servers (GPU instances)
├── Database Cluster (Primary + 2 replicas)
├── Redis Cluster (3 nodes)
└── S3-compatible Storage (media files)
```

CI/CD Pipeline

```
GitHub → GitHub Actions → Build Docker Image →
Push to Registry → Deploy to Kubernetes →
Run Tests → Blue-Green Deployment → Production
```

Success Metrics & KPIs

Business Metrics

- **Monthly Recurring Revenue (MRR)**
- **Churn rate** (target: < 10%)
- **Customer Lifetime Value (LTV)** (target: > $200)
- **Daily/Monthly Active Users (DAU/MAU)**
- **Conversion rate** (free to paid, target: 2-5%)

Technical Metrics

- **API response times** (target: < 200ms)
- **Error rates** (target: < 1%)
- **Image generation queue depth**
- **Database query performance**
- **Uptime** (target: 99.9%)

Growth Targets

Month 1 (Launch)

- **Target:** 100 signups, 5 paying subscribers
- **Revenue:** $125
- **Status:** Breaking even on infrastructure

Month 3

- **Target:** 500 signups, 50 paying subscribers
- **Revenue:** $1,250
- **Status:** Covering basic costs

Month 6

- **Target:** 2,000 signups, 200 paying subscribers
- **Revenue:** $5,000
- **Status:** Profitable (minimal setup)

Month 12

- **Target:** 10,000 signups, 1,000 paying subscribers
- **Revenue:** $25,000
- **Status:** Scaling rapidly

Risk Mitigation

Technical Risks

Risk	Probability	Impact	Mitigation
Payment gateway ban	Medium	Critical	Multi-gateway orchestration, crypto backup
AI model failure	Low	High	Fallback models, monitoring
Server downtime	Medium	High	Auto-scaling, multi-region
Data breach	Low	Critical	Encryption, security audits

Business Risks

Risk	Probability	Impact	Mitigation
Low user acquisition	High	Critical	Marketing budget, SEO, referrals
High churn rate	Medium	High	Better UX, personalization

Risk	Probability	Impact	Mitigation
Legal issues	Low	Critical	Lawyer consultation, compliance
Competition	High	Medium	Unique features, better UX

Conclusion

The IMPS platform requires careful consideration of:

1. **Payment processing** - Adult-specific gateways with orchestration backup
2. **Compliance** - Mandatory age verification, content moderation
3. **Security** - End-to-end encryption, data protection
4. **Scalability** - Microservices architecture, GPU clusters for AI
5. **Legal** - Proper business structure, comprehensive policies

Key Success Factors

✅ Start with MVP (don't over-engineer)
✅ Focus on user experience
✅ Ensure payment stability (multi-gateway)
✅ Market aggressively
✅ Listen to user feedback

Summary Metrics

- **Time to Launch MVP:** 3-4 months
- **Initial Investment:** $15,000-30,000
- **Monthly Operating Costs:** $3,000-5,000 (minimal) to $30,000+ (full-scale)
- **Breakeven Point:** 133 paying users
- **Target Market Size:** Rapidly growing ($10M+/year for top platforms)

With proper execution and the roadmap outlined in this document, IMPS has strong potential to capture market share in the AI companion industry.

Document Version: 1.0
Last Updated: October 17, 2025
Author: Jaguo415
Repository: https://github.com/Jaguo415/imps-architecture-docs